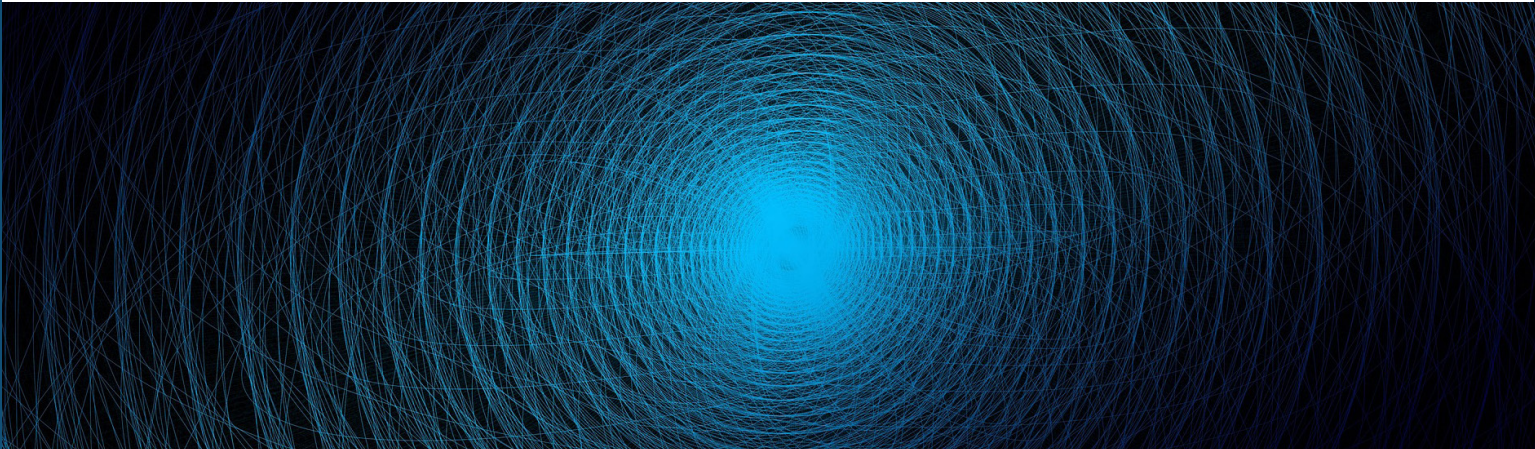




JAPAN'S NEXT MOONSHOT: LEADERSHIP IN HIGH-PERFORMANCE COMPUTING COUPLED WITH QUANTUM COMPUTING TECHNOLOGY

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Through public-private collaboration, Japan is pushing the technology frontier in next-generation high-performance computing (HPC) and quantum computing, beyond current-generation artificial intelligence (AI). Sizable investments are beginning to yield results, from building the infrastructure for next-generation hardware to new start-ups. This new moonshot leverages Japan's long-standing core competencies in high-level precision manufacturing, hardware excellence, system-level engineering, computing design innovation, and industrial policy. It is also a central piece in Japan's response to the rise of geoeconomics. Rather than viewing the world in only zero-sum terms, Japan's positioning begins with a conviction that nobody can win alone in the new world of HPC and quantum computing. To balance power relations in the U.S.-Japan security umbrella, Japan's national economic security strategy is centered on forging inextricable tie-ups with the United States.

Japan's New Moonshot: Quantum and High-Performance Computing

AI is on everybody's mind, in Japan just as much as everywhere else. While Japan may have been late in developing its own AI, there is now substantial investment and activity around domestic AI design. These are often specialized to certain areas; for example, the NEC cotomi focuses on helping companies accelerate their digital transformation. In October 2025 a new government push was announced to build more domestic AI in order to reduce dependency on U.S. and Chinese AI models.¹

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In addition to these activities related to current-generation AI, Japan has launched a separate so-called moonshot goal for quantum computing, and is making similar efforts to advance HPC capabilities. Perhaps different from the United States, where the concept of AI is sometimes used as an umbrella term that encompasses a vast range of next-generation computing technologies, stakeholders in the Japanese private and public sectors consider HPC separate from AI and look to tie HPC to quantum computing. The government's ten-point Moonshot R&D Program includes AI and robotics as the third goal and quantum as a separate, sixth goal. In this broader ecosystem, HPC is seen as a technology that could utilize quantum hardware and middleware in the event of a quantum computing breakthrough.²

Put in highly simplistic terms, the current AI is a black box, a closed system that identifies patterns in existing information to make predictions. This is most useful for deep learning, model training, and data inference. According to some experts, such as Yann LeCun, AI is not an end state but an enabling technology. It is most effective when used as a tool to do other things, such as writing software or solving scientific problems. LeCun has also predicted that more advanced systems will replace AI based on large language models in rapid succession.

In contrast, HPC can be thought of as a white box, a kind of computing architecture designed to solve complex scientific problems on its own. Rather than processing massive amounts of data as fast as possible for fast learning, as AI does, in HPC every node must talk to its neighbor instantly in the pursuit of very high precision and exact accuracy. Some might think of HPC as the backbone to AI, while others view AI as the accelerator for HPC.³ In a pathbreaking event, in 2025 the Japanese start-up Sakana AI managed to have its "AI scientist" generate a peer-reviewed journal publication. In other words, the company produced a completely machine-generated research paper that presented new scientific discovery and passed the journal review process.⁴

Japan's quantum moonshot is about leading the development of advanced hardware, middleware, and software systems that could support future HPC applications. The push in this direction began under the heading of AI, including in the passage of the AI Promotion Act of 2025 and the establishment of the AI Strategic Headquarters in the prime minister's office.⁵ Soon thereafter, Japan's Ministry of Economy, Trade and Industry (METI) broadened the goal and labeled the year 2025 as the "first year of quantum industrialization," complete with a "high-stakes, preemptive" government intervention exceeding \$7 billion.⁶ Since then, public-private efforts in this next-generation area have skyrocketed.

For the five years from 2025 to 2030, over 10 trillion yen (\$65 billion, which is more than the amount authorized in the U.S. CHIPS and Science Act) have been earmarked for government investment in infrastructure upgrades (such as submarine cable networks) and next-generation chip and quantum-computing research (such as the Rapidus next-generation semiconductor production facility in Hokkaido) that Japanese stakeholders believe will serve as a foundation for HPC. The government initiative has also earmarked funds for start-up companies, such as Sakura Internet, a Gen-AI platform based on NEC's cotomi-AI for advanced cloud-based supercomputing applications.⁷

The efforts to spearhead quantum-computing powers are led by RIKEN, Japan's preeminent national research laboratory. In collaboration with Fujitsu and Nvidia, RIKEN has developed FugakuNEXT, which is the quantum-computing hardware, as well as OQTOPUS, a quantum-computing software system developed by RIKEN and Fujitsu. RIKEN is also supporting a SoftBank initiative to create a platform for quantum-computing research that involves 21 companies and research teams. Meanwhile, the ABCI-Q Supercomputer is developed jointly between the National Institute of Advanced Industrial Science and Technology (a METI-affiliated research agency) and SoftBank.⁸ METI has also created a

new facility called G-QuAT to serve as a hub for business collaboration to foster talent and speed up technology transfer.

These programs are reminiscent of the research consortia of the 1980s, and indeed Japan's long experience in public-private R&D collaboration is affording the country once again a great advantage. With the goal of bringing together corporate R&D leaders and bridging academic research and commercial application, METI and RIKEN are building an ecosystem that can turn scientific progress into national competitiveness.⁹

The above developments represent the highlights rather than the entire spectrum of Japan's activities at the intersection of AI, HPC, and quantum computing. Moreover, Japan's ambitions are not solely focused on building high-powered hardware. Investments are also directed at middleware—i.e., the software that allows quantum devices to run applications. For example, the telecom company KDDI and start-up Jiji are building a first quantum operating system that, if successful, could secure Japan a dominant position in a potential future quantum ecosystem.

Japan's Business Reinvention and HPC Leadership

The government initiatives are complemented by private investment that already exceeds \$70 billion, if it can even be fully assessed. The results of the IT industry's massive efforts are now beginning to show, in ways that may be surprising for a casual observer who has not looked at Japan's IT industry since the bubble economy burst in the early 1990s. The former "NTT family" members—in particular, NEC, Fujitsu, and Hitachi—have awakened from their slumber and found their own new meaning and energy in the new Moonshot Program. Each of them is undertaking its own digital transformation and building AI platforms as well as hardware, with investments well over \$30 billion so far. SoftBank, KDDI, and others have joined the club, with SoftBank alone investing \$49 billion in the Stargate project with OpenAI. The

NTT Group itself has pledged \$59 billion to build the new Smart AI Agent Ecosystem. This frantic activity has attracted foreign investments as well, including by Microsoft, Amazon Web Services, and Alphabet, which have pledged over \$15 billion to upgrade their cloud infrastructure in Japan.¹⁰

This wave of investment is happening as a critical mass of Japan's leading companies are making rapid progress in a decade-long business reinvention. These frontrunners have refocused their core competencies away from consumer-end products and into the technologically advanced upstream and downstream portions of global value chains. Because this new expertise is quietly building in the high-upstream portions of global value chains, such as in advanced materials or machinery, it is often overlooked. Nevertheless, in many areas Japan's leading companies have become key players in various areas of the technological frontier.¹¹

As a result of this shift upstream and toward the technology frontier, Japan overall has moved from a "Made in Japan" business model to a "Japan Inside" one.¹² The new leadership is currently most evident in advanced materials and components that feed into the making of many products that do not carry a Japanese brand name. Yet, this has made Japanese companies integral to many supply chains, including in next-generation computing, data centers, and smart cities.

Japan's Moonshot Program is leveraging these new strengths. While U.S. firms are focusing their investments on AI in consumer-facing products, Japan is positioning upstream, in the development of hardware and middleware (system-engineering) applications. With these efforts, Japan is angling to become an indispensable collaborator in the next generation of HPC and quantum hardware, middleware, and software.

Collaboration, Not Just Cooperation

The global shift toward geoeconomics—using economic policy as a tool of foreign policy and

for strategic gain—plays into Japan’s hands given long-standing strong government-business ties. As the world enters a new order where countries no longer pursue shared prosperity but rather national economic gains in purportedly zero-sum games, it is turning away from free-market competition and toward corporate strategy guided and perhaps even governed by national security interests.¹³

In this new world of geoeconomics and economically driven national security strategies, one might think that Japan is entering the ring from a position of weakness. Japan greatly depends on the United States for defense in Northeast Asia, and everything in Japanese policymaking today must be viewed in the context of its efforts at maintaining the U.S.-Japan security umbrella.

However, to balance the dependency, Japan’s strategy is to make itself an indispensable partner to as many countries and companies as possible. The path Japan has chosen toward this goal is collaboration, even as the rest of the world is turning toward isolation. The quantum moonshot is a great example of this strategy of building crucial bargaining chips that make Japan all but indispensable at the technology frontier.

Japan begins this technology pursuit with the assumption that no one can win the race in HPC or quantum computing alone. Domestically, this justifies concerted government-industry efforts at R&D, relying on time-tested playbooks of research consortia and government-business collaboration. Globally, Japan aims to be the reliable, go-to partner for the United States and other collaborators by proactively seeking commitments, joint investments,

and corporate and personal tie-ups. In some circles in Japan, the term “cooperation” is considered too weak for these partnerships, and in many official statements Japan now explicitly insists on a deeper, more inextricable form of “collaboration.”¹⁴

This approach fundamentally differs from the U.S. approach. The U.S. AI Action Plan sets as the explicit goal “technological dominance,” including pushing the United States’ AI vision and ecosystem globally.¹⁵ In contrast, Japan’s official stance is to seek “technological leadership,” which implies a more collaborative approach and aims to make the country one of the star players in a team effort.¹⁶ In other words, Japan’s moonshot programs are intended to increase its indispensability.

In October 2025 the United States and Japan signed a memorandum of cooperation that Japan would make available \$550 billion of funds for investment in the United States.¹⁷ While the details will most likely be revised repeatedly, this memorandum could become a centerpiece in U.S.-Japan collaboration in HPC. As the exact investment areas are negotiated, Japan’s goal remains constant, namely to create synergies with the United States.

In sum, Japan is building world-class infrastructure for HPC and quantum computing, with substantial government commitment of resources and renewed activity in the private sector to position the country for collaboration in a world of geoeconomics. This collaborative stance makes Japan an exception in an era when many countries are engaged in zero-sum competition. ∞

Endnotes

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